THE COTANGENT COMPLEX OF A PRODUCT OF ALGEBRAS IS THE SUM OF THE COTANGENT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author explicitly chooses the direct elementary proof over the deduction from the fundamental triangle; both the choice and the argument are inside the counted span.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The special case of the fundamental triangle for a polynomial extension. Lemma lemma-special-case-triangle, used at proof lines 1288-1302. Produces the three distinguished triangles used in the reduction.
- Flat base change for the cotangent complex. Lemma lemma-flat-base-change-cotangent-complex, used at proof lines 1304-1310. Identifies the cotangent complexes after inverting x.
- The factorization A -> A[x] -> B x C with x mapping to (1, 0). used at proof lines 1288-1289. Supplies the polynomial extension whose localizations separate B and C.
- Vanishing of C tensor_{A[x]} A[x, x^{-1}] and the identification of the localization of B x C. used at proof lines 1305-1307. Make the two localizations reduce to the known cases.

Source reference. Original lemma 08QQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11068–11068. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08SA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11123–11123. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A \to B$ and $A \to C$ be ring maps. Then the map $L_{B \times C/A} \to L_{B/A} \oplus L_{C/A}$ is an isomorphism in $D(B \times C)$.*

Proof. Although this lemma can be deduced from the fundamental triangle we will give a direct and elementary proof of this now. Factor the ring map $A \to B \times C$

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as $A \to A[x] \to B \times C$ where $x \mapsto (1, 0)$. By Lemma 08SA we have a distinguished triangle

$$L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} (B \times C) \to L_{B\times C/A} \to L_{B\times C/A[x]} \to L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} (B \times C)[1]$$

in $D(B \times C)$. Similarly we have the distinguished triangles

$$L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} B \to L_{B/A} \to L_{B/A[x]} \to L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} B[1]$$
$$L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} C \to L_{C/A} \to L_{C/A[x]} \to L_{A[x]/A} \otimes^{\mathbf{L}}_{A[x]} C[1]$$

Thus it suffices to prove the result for $B\times C$ over $A[x]$. Note that $A[x] \to A[x, x^{-1}]$ is flat, that $(B\times C)\otimes_{A[x]} A[x, x^{-1}] = B\otimes_{A[x]} A[x, x^{-1}]$, and that $C\otimes_{A[x]} A[x, x^{-1}] = 0$. Thus by base change (Lemma 08QQ) the map $L_{B\times C/A[x]} \to L_{B/A[x]} \oplus L_{C/A[x]}$ becomes an isomorphism after inverting x. In the same way one shows that the map becomes an isomorphism after inverting $x - 1$. This proves the lemma. $\square$